1.6 Matrix Elements of a Linear Operator

Let $\{|i\rangle\}_{i=1}^{n}$ be a complete orthonormal basis, and let $|\psi\rangle = \Omega|v\rangle$. Expanding each state in this basis gives

$$|\psi\rangle = \sum_i |i\rangle\langle i \mid \psi\rangle, \qquad |v\rangle = \sum_j |j\rangle\langle j \mid v\rangle.$$

Expanding the matrix element

Substituting these expansions into $\langle w|\Omega|v\rangle$, we obtain

$$\begin{aligned}
\langle w|\Omega|v\rangle &= \langle w \mid \psi\rangle \\
&= \langle w|\Big(\sum_i |i\rangle\langle i \mid \psi\rangle\Big) \\
&= \sum_i \langle w \mid i\rangle\langle i \mid \psi\rangle \\
&= \sum_i \langle w \mid i\rangle\langle i|\Omega|v\rangle \\
&= \sum_i \langle w \mid i\rangle\langle i|\Omega\Big(\sum_j |j\rangle\langle j \mid v\rangle\Big) \\
&= \sum_{i,j} \langle w \mid i\rangle\langle i|\Omega|j\rangle\langle j \mid v\rangle.
\end{aligned}$$

Matrix representation

The quantities $\Omega_{ij} = \langle i|\Omega|j\rangle$ are the matrix elements of Ω in the chosen basis. The same expression is a matrix product:

$$\langle w|\Omega|v\rangle = \begin{bmatrix} \langle w \mid 1\rangle \\ \langle w \mid 2\rangle \\ \vdots \\ \langle w \mid n\rangle \end{bmatrix}^T \begin{bmatrix} \langle 1|\Omega|1\rangle & \langle 1|\Omega|2\rangle & \cdots & \langle 1|\Omega|n\rangle \\ \langle 2|\Omega|1\rangle & \langle 2|\Omega|2\rangle & \cdots & \langle 2|\Omega|n\rangle \\ \vdots & \vdots & \ddots & \vdots \\ \langle n|\Omega|1\rangle & \langle n|\Omega|2\rangle & \cdots & \langle n|\Omega|n\rangle \end{bmatrix} \begin{bmatrix} \langle 1 \mid v\rangle \\ \langle 2 \mid v\rangle \\ \vdots \\ \langle n \mid v\rangle \end{bmatrix}.$$

Here T denotes the ordinary transpose: the entries $\langle w \mid i\rangle$ already satisfy $\langle w \mid i\rangle = \langle i \mid w\rangle^*$.

Completeness: the resolution of the identity

The basis expansion of an arbitrary state implies

$$\boxed{\sum_i |i\rangle\langle i| = I.}$$

Thus the double sum can also be obtained by inserting the identity on both sides of the operator:

$$\begin{aligned}
\langle w|\Omega|v\rangle &= \langle w|\Big(\sum_i |i\rangle\langle i|\Big)\Omega\Big(\sum_j |j\rangle\langle j|\Big)|v\rangle \\
&= \sum_{i,j} \langle w \mid i\rangle\langle i|\Omega|j\rangle\langle j \mid v\rangle.
\end{aligned}$$